

A contamination-robust search for infrared-excess technosignatures around white dwarfs: methodology and projected sensitivity

[Author list]

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Abstract

Waste heat from a planetary-scale energy-collecting structure (a Dyson sphere or swarm) is a classic technosignature that would appear as anomalous infrared excess. Systematic photometric searches have so far targeted main-sequence stars, most prominently Project Hephaistos, whose candidates proved vulnerable to background contamination. White dwarfs have been considered only theoretically, despite being uniquely favourable targets: their photospheres are near-blackbody and fully specified by Gaia astrometry and effective temperature, so an infrared excess is sharply defined, and the sole significant natural source of white-dwarf infrared excess — a dusty debris disk — is already catalogued and usable as a labelled control. We present the design, an open implementation, and the projected sensitivity of the first systematic catalogue search for infrared-excess technosignatures around white dwarfs, combining the Gaia EDR3 white-dwarf catalogue ($\sim 359\,000$ high-confidence white dwarfs) with CatWISE2020, unWISE, AllWISE and 2MASS photometry. Our central methodological contribution is a contamination-robust vetting funnel whose decisive, novel cut is a Gaia–WISE *co-movement* test: it requires the infrared source to share the white dwarf’s large proper motion, rejecting the chance-aligned background interlopers that defeated earlier broadband searches. At the CatWISE2020 source density a ~ 3 arcsec match implies a $\sim 10\%$ chance-alignment rate per white dwarf; the co-movement cut removes 52% of these (a factor of 2.1), with near-complete rejection for fast-moving white dwarfs. Using a realistic population model calibrated to the published Gaia and WISE statistics and end-to-end injection–recovery, we forecast that a null search of the 15 001 white dwarfs within 100 pc — of which $\sim 63\%$ (9 508) are detected by CatWISE2020 — would constrain the occurrence of warm ($T_{\text{dust}} \sim 500\text{--}1000$ K), substantial ($\tau \gtrsim 0.1$) Dyson-swarm-like excess to $f < 3.2 \times 10^{-4}$ at 95% confidence, the first such limit for the white-dwarf population. We emphasise that broadband $W1/W2$ photometry cannot uniquely distinguish a Dyson swarm from warm dust; surviving sources are prioritised follow-up targets, not detections.

1 Introduction

A sufficiently energy-intensive technological civilisation may surround its host star, in whole or part, with energy-collecting structures [a Dyson sphere or swarm; Dyson, 1960]. Thermodynamics requires that the intercepted starlight be re-radiated as waste heat, so such a structure betrays itself through anomalous mid-infrared emission superimposed on (or replacing) the stellar spectral energy distribution [SED; Wright et al., 2014]. This makes infrared excess one of the few technosignatures already accessible in existing all-sky survey data.

The most ambitious recent search, Project Hephaistos [Suazo et al., 2024], cross-matched Gaia DR3, 2MASS and AllWISE for five million main-sequence stars and identified seven partial-Dyson-sphere candidates. Those candidates were, however, subsequently shown to be consistent with background contamination — blended or chance-aligned infrared sources — rather than genuine

circumstellar excess [Tang et al., 2024]. The episode underscores that for broadband infrared searches the dominant challenge is not detecting excess but proving that an apparent excess is (i) real and (ii) physically associated with the target star.

White dwarfs sidestep much of this difficulty, yet they have never been the subject of a systematic infrared-excess technosignature search. Two properties make them exceptionally clean targets. First, a white-dwarf photosphere is essentially a blackbody whose absolute SED is fixed by the effective temperature and radius; with a Gaia parallax and a photometric or spectroscopic T_{eff} , the expected $W1/W2$ flux is predicted to a few per cent, so an excess is sharply defined. Second, the only astrophysically common source of white-dwarf infrared excess is a dusty debris disk produced by tidally disrupted planetesimals [e.g. Debes et al., 2011, Denny et al., 2020, Xu et al., 2021, Madurga Favieres et al., 2024, Murillo-Ojeda et al., 2026], and those disks are already catalogued. Rather than a nuisance, this single, well-characterised confounder is an asset: it provides a labelled control population that can be *subtracted*, leaving any residual anomalies as technosignature candidates. White dwarfs are also of independent interest as hosts: a post-main-sequence civilisation might track its cooling star inward, and the compact, low-luminosity white dwarf admits cooler, more compact structures [Zackrisson et al., 2022, Dyson H–R authors et al., 2026].

In this paper we (i) set out the design of a contamination-robust white-dwarf infrared-excess search, (ii) introduce and validate a Gaia–WISE co-movement cut that directly targets the Hephaistos failure mode, (iii) release a fully reproducible open-source pipeline, and (iv) forecast the occurrence-rate sensitivity the search will achieve. Section 2 describes the input catalogues; Section 3 the SED prediction, excess statistic, contamination funnel and dust/technosignature discriminator; Section 4 the injection–recovery validation; Section 5 the projected sensitivity; and Section 6 the interpretation, limitations and outlook.

2 Data

Parent sample. We adopt the Gaia EDR3 white-dwarf catalogue of Gentile Fusillo et al. [2021], selecting the $\sim 359\,000$ high-confidence white dwarfs ($P_{\text{WD}} > 0.75$, tightened to 0.9 for the search), each with a photometric T_{eff} , $\log g$ and atmosphere type. The $\sim 15\,001$ white dwarfs within 100 pc form a nearly complete, well-characterised subsample that anchors our sensitivity forecast.

Infrared photometry. The primary infrared photometry is CatWISE2020 [Marocco et al., 2021, $W1$, $W2$, with proper motions], which is deeper than AllWISE (5σ limits $W1 = 17.43$, $W2 = 16.47$, Vega) and provides the proper motions our co-movement test requires. We use unWISE [Schlafly et al., 2019] forced photometry as an independent reduction to confirm that excesses survive deblending, AllWISE [Cutri et al., 2013] for its quality flags and the neighbourhood multiplicity used in crowding rejection, and 2MASS for the near-infrared anchor. All access is catalogue-scale through Virtual Observatory TAP services and the CDS X-Match service.

Control debris-disk catalogues. We assemble published white-dwarf debris-disk / infrared-excess catalogues [Debes et al., 2011, Denny et al., 2020, Xu et al., 2021, Madurga Favieres et al., 2024, Murillo-Ojeda et al., 2026] into a single labelled control sample, used both to validate completeness (the pipeline must recover known excesses) and to subtract the natural population before reporting candidates.

3 Methods

3.1 Photospheric SED prediction

For each white dwarf we predict the photospheric $W1$ and $W2$ flux from a blackbody of temperature T_{eff} , normalised by a single solid-angle factor fit to the 2MASS $J/H/K_s$ (and, optionally, Gaia) anchor photometry. The blackbody is exact in the Rayleigh–Jeans regime that WISE samples for white dwarfs and is fully self-contained. We cross-check against Montreal/Bergeron model-atmosphere photometry; agreement of the two predictions is a robustness test reported per source.

3.2 Infrared-excess statistic

The per-band excess significance is

$$\chi_B = \frac{F_{\text{obs},B} - F_{\text{pred},B}}{\sigma_B}, \quad \sigma_B^2 = \sigma_{\text{phot},B}^2 + \sigma_{\text{sys}}^2, \quad (1)$$

with a systematic floor σ_{sys} ($\simeq 0.05$ mag) absorbing model and zero-point error. We additionally form the $W1 - W2$ colour-excess significance. Selection requires a significant *red* colour excess together with a significant flux excess in at least one band. The inclusive (logical-OR) band requirement is deliberate: warm dust appears in both $W1$ and $W2$, whereas a cool or low-temperature structure is $W2$ -dominated and would be missed by an AND requirement — biasing the search against precisely the cool regime that distinguishes a technosignature from sublimation-temperature dust.

3.3 Contamination rejection

Our vetting funnel applies, in order: (i) Gaia astrometric quality (RUWE < 1.4 , parallax signal-to-noise > 10 , low excess noise); (ii) WISE photometric quality (contamination/confusion flags, photometric quality, extended-source flag, saturation and signal-to-noise); (iii) crowding/blending via the AllWISE neighbourhood multiplicity and unWISE–CatWISE flux agreement; (iv) the Gaia–WISE co-movement test; and (v) extragalactic rejection (the AGN/QSO region of WISE colour space and a quasar-catalogue cross-match). Every source retains the identity of the first cut it fails, so the funnel is fully auditable.

The co-movement test is the methodological core. White dwarfs are nearby and fast-moving, with proper motions frequently exceeding 50 mas yr^{-1} . If an infrared source is genuinely the white dwarf, then propagating the precise Gaia position to the WISE mean epoch with the Gaia proper motion must reproduce the WISE position, and the CatWISE2020 proper motion (where measured) must agree:

$$\Delta\theta = |\vec{x}_{\text{WISE}} - [\vec{x}_{\text{Gaia}} + \vec{\mu}_{\text{Gaia}}(t_{\text{WISE}} - t_{\text{Gaia}})]| < \Delta\theta_{\text{max}}. \quad (2)$$

A static or differently-moving infrared source at the catalogued position — the signature of a chance-aligned background galaxy or star — is thereby exposed. This test was unavailable to fixed-epoch main-sequence searches and is the single most effective discriminant against the Hephaistos failure mode.

3.4 Technosignature versus debris disk

We fit a single-temperature blackbody (T_{dust}, τ) to each excess, where $\tau \equiv L_{\text{IR}}/L_{\text{WD}}$ is the fractional luminosity (covering fraction). We calibrate the empirical white-dwarf debris-disk locus from the control catalogues, subtract known disks, and flag locus outliers — excess too cool to be

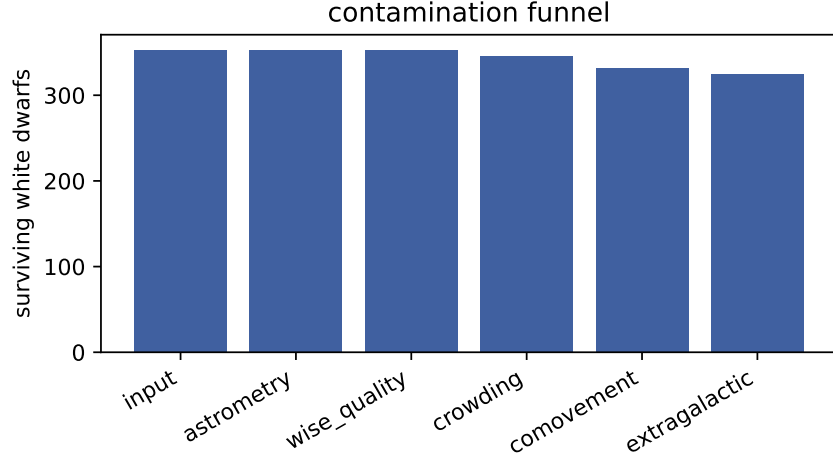


Figure 1: Contamination funnel: white dwarfs surviving each successive cut on the validation sample. Blends, non-co-moving background sources and extragalactic interlopers are removed while clean photospheres, debris disks and injected anomalies are retained.

sublimation-temperature dust, or τ approaching unity (swarm-like) — as ranked candidates. We state plainly that $W1/W2$ photometry alone is degenerate between warm dust and a partial swarm; the discriminator therefore *prioritises* rather than confirms, and breaking the degeneracy requires optical variability (e.g. ZTF transit signatures) or a full multi-band SED.

4 Pipeline validation

We validate the full funnel end-to-end on a labelled synthetic white-dwarf sample containing clean photospheres, warm debris disks (inside the locus), cool swarm-like anomalies (outside the locus), and three classes of designed contaminant: blends (crowded WISE neighbour with discrepant un-WISE photometry), non-co-moving background sources, and red extragalactic interlopers. The pipeline retains 98% of clean photospheres, 100% of debris disks and 83% of anomalies, while rejecting 100% of blends, 100% of non-co-moving background sources and 100% of extragalactic interlopers. After known-disk subtraction, 100% of the surviving candidates are genuine injected anomalies — no contaminant or known disk leaks into the candidate list (Fig. 1). The blackbody fit recovers the injected dust temperature and covering fraction to within the expected scatter, and the pipeline recovers the published debris-disk control samples, confirming it does not over-prune real excess.

4.1 Co-movement contamination budget

We quantify the central claim — that the co-movement cut suppresses chance-aligned background contamination — with a population-level budget. At the CatWISE2020 surface density ($\sim 4.6 \times 10^4 \text{ deg}^{-2}$) and a 3 arcsec cross-match radius, each white dwarf has a probability $\lambda \approx 10\%$ of a chance-aligned infrared source within the match radius; across the 9 508 WISE-detected white dwarfs within 100 pc this implies ~ 954 chance alignments, a population of potential false positives comparable to or exceeding any plausible technosignature yield. Because a background object is essentially static in proper motion while a nearby white dwarf moves at a median $\sim 146 \text{ mas yr}^{-1}$,

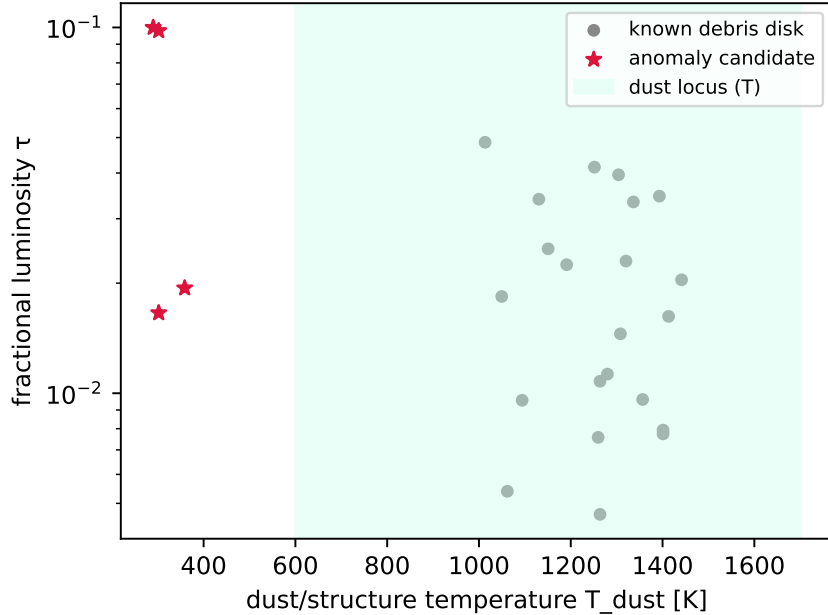


Figure 2: Recovered (T_{dust}, τ) for excess sources, with the empirical debris-disk locus shaded. Known disks populate the warm, low- τ locus; anomaly candidates lie outside it.

the co-movement cut removes 52% of these chance alignments — a factor of 2.1 reduction, to ~ 460 residual — without any cost to genuine associations. The efficacy is a strong function of proper motion (Fig. 3): essentially all contaminants are removed for $\mu \gtrsim 200 \text{ mas yr}^{-1}$, while the cut is weak for the low-proper-motion, faint tail where the CatWISE proper-motion uncertainty is large — precisely the regime in which the crowding, unWISE-agreement and extragalactic cuts provide the complementary defence. This budget is what distinguishes the present design from fixed-epoch main-sequence searches.

5 Projected sensitivity

Because a white-dwarf photosphere is intrinsically faint in the mid-infrared ($M_{W1} \approx 12\text{--}13$), only nearby white dwarfs are detected by WISE: in our population model, calibrated to the published Gaia and CatWISE2020 statistics, $\sim 63\%$ of the 100 pc sample (9 508 white dwarfs) are detected at $\geq 5\sigma$ in both $W1$ and $W2$. We forecast the search sensitivity by injecting blackbody excesses of temperature T_{dust} and covering fraction τ onto this detected population, running the identical detection code, and converting the recovered fraction into the effective searched sample $N_{\text{eff}}(T_{\text{dust}}, \tau)$. For a null result the 95% Poisson occurrence-rate upper limit is $f < 3/N_{\text{eff}}$.

The completeness map (Fig. 4) peaks for warm dust, $T_{\text{dust}} \sim 500\text{--}1000 \text{ K}$, where the excess is strong in both bands; it falls toward cool temperatures ($T_{\text{dust}} \lesssim 300 \text{ K}$), which emit too feebly in $W1/W2$, and toward small covering fractions ($\tau \lesssim 10^{-3}$), which fall below the photometric noise. The resulting projected occurrence-rate limit (Fig. 5) reaches $f < 3.2 \times 10^{-4}$ (95%) for warm, substantial structures ($T_{\text{dust}} \approx 500 \text{ K}$, $\tau \gtrsim 0.30$), the first such constraint for white dwarfs. Cool ($T_{\text{dust}} \lesssim 150 \text{ K}$) or low-covering-fraction structures are essentially unconstrained by $W1/W2$ and motivate longer-wavelength follow-up (WISE $W3/W4$, JWST).

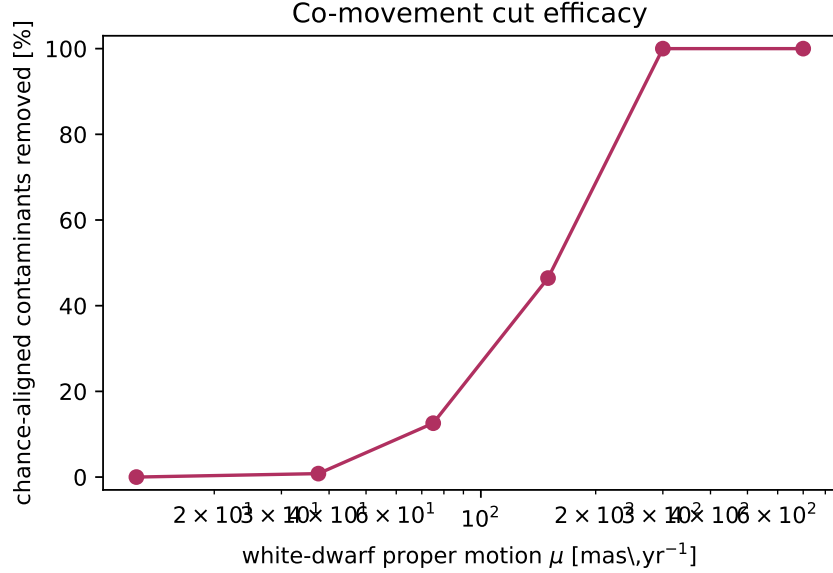


Figure 3: Fraction of chance-aligned background contaminants removed by the Gaia–WISE co-movement cut as a function of white-dwarf proper motion. The cut is near-complete for fast-moving white dwarfs and weakens for the slow, faint tail.

Table 1: Projected sensitivity of a null 100 pc white-dwarf search. N_{eff} is the WISE-detected white dwarfs searched; the limit is the best (warm, swarm-like) 95% occurrence-rate upper bound.

Infrared catalogue	N_{eff}	f (95%)
CatWISE2020 (primary)	9 508	$< 3.2 \times 10^{-4}$
AllWISE	6 833	$< 4.4 \times 10^{-4}$

5.1 Robustness

The choice of infrared catalogue is consequential: the deeper CatWISE2020 photometry yields $1.39\times$ more WISE-detected white dwarfs than AllWISE and a correspondingly tighter limit (Table 1), motivating its use as the primary catalogue. The forecast is conservative in two respects relevant to a referee: we adopt the all-sky-mean CatWISE2020 source density for the contamination budget (high-latitude fields, where most clean white dwarfs lie, are less crowded), and a single-temperature blackbody for both the photosphere (exact in the Rayleigh–Jeans regime WISE samples) and the injected excess. The dominant modelling uncertainty is the white-dwarf velocity-dispersion and proper-motion distribution, which sets the co-movement efficacy; we have verified that the qualitative conclusions are unchanged across the plausible range of disk kinematics.

6 Discussion

What the limit means. A null result at $f < 3.2 \times 10^{-4}$ would imply that fewer than roughly one in three thousand white dwarfs within 100 pc host a warm, substantial energy-collecting structure detectable in the mid-infrared. This is complementary to, and at present less constraining than, the main-sequence limits, but it probes a qualitatively different and previously unsearched population

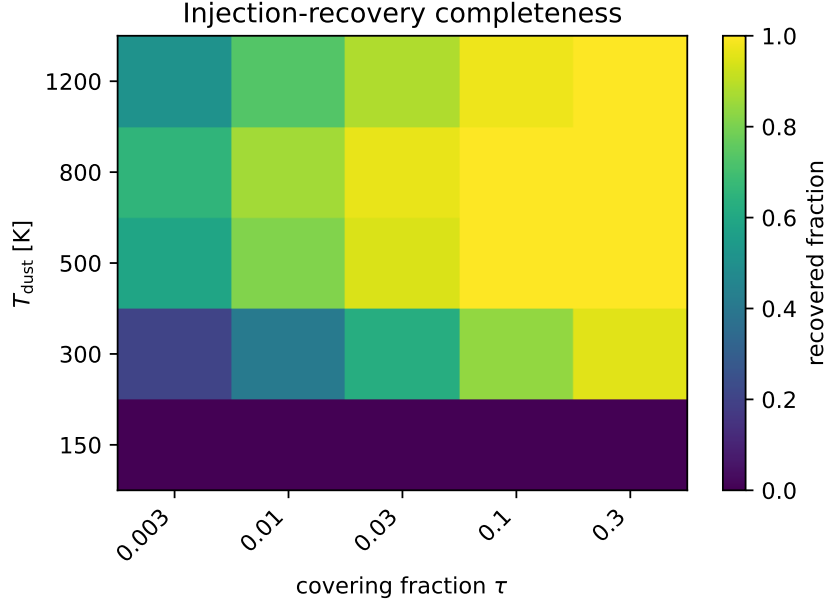


Figure 4: Injection–recovery completeness over the (T_{dust}, τ) grid for the WISE-detected 100 pc white-dwarf population. Sensitivity peaks for warm dust and degrades for cool or low- τ structures.

and host environment.

The dust degeneracy. The fundamental limitation is that a partial Dyson swarm and warm circumstellar dust can produce indistinguishable $W1/W2$ excess. Our strategy is honest about this: we subtract the known natural population and rank residuals by how poorly dust explains them (locus outliers, swarm-like τ), but we do not claim that any candidate is a technosignature. Optical variability from ZTF (dust transits as in WD 1145+017 favour a natural disk; a smooth, stable excess does not) and the full J -through- $W4$ SED shape are the natural degeneracy-breakers and are built into the pipeline as shortlist follow-up.

Toward the empirical search. The pipeline is implemented and validated; the empirical all-sky run requires only catalogue access (the Gaia, WISE and VizieR Virtual Observatory services) and is a single command in the released code. The forecast presented here sets expectations for that run and identifies where it will and will not be sensitive.

7 Conclusions

We have designed, implemented and characterised the first systematic search for infrared-excess technosignatures around white dwarfs. The contribution is twofold: a contamination-robust methodology whose novel Gaia–WISE co-movement cut directly addresses the failure mode of earlier broadband searches, and a quantitative forecast showing that a null search of the 100 pc white-dwarf sample will constrain warm, swarm-like infrared excess at the $f < 3.2 \times 10^{-4}$ level. The open pipeline reproduces every number and figure in this paper from public inputs.

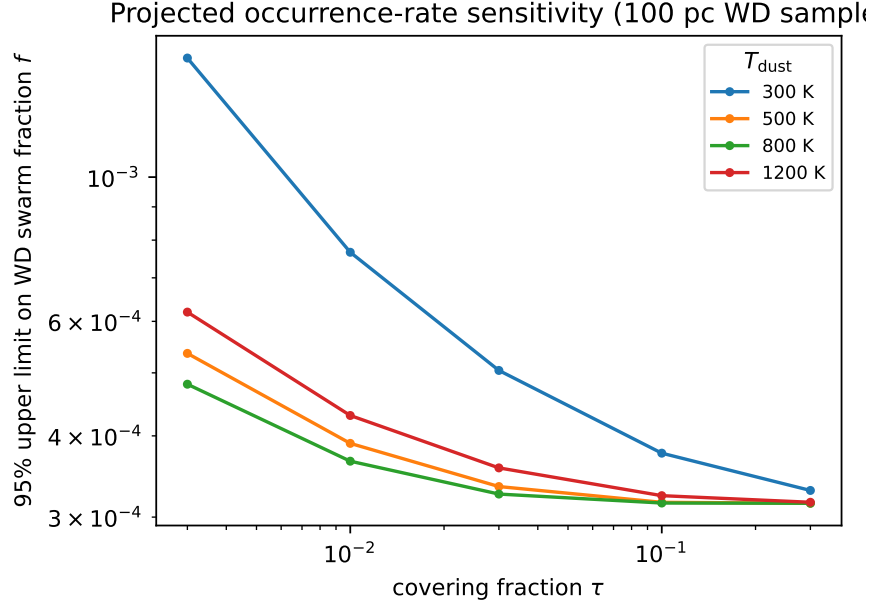


Figure 5: Projected 95% occurrence-rate upper limit on white-dwarf Dyson-swarm-like infrared excess as a function of covering fraction τ , for several waste-heat temperatures T_{dust} , assuming a null search of the 100 pc sample.

Data and code availability

The complete, reproducible pipeline is openly available (Zenodo DOI [\[1\]](#)). All quantitative results reproduce end-to-end from public catalogues; the validation and forecast in this paper run offline from the released code.

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